From Data to Decisions:

Interactive Dashboards for Museum Engagement

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Background

The Ann Arbor Hands-On Museum (AAHOM) is a nonprofit serving 300,000+ learners annually across its Kerrytown and Leslie Nature Center locations. Their mission is to create moments of discovery that inspire curiosity for STEAM learning and exploration.

Museum Exhibits

Preschool Gallery

Steam Park

H_2'Oh!' Water Exhibit

Problem Statement

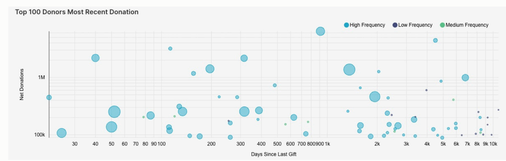
The Ann Arbor Hands-On Museum would like to enhance its tracking of YoY attendance through deeper segmentation across visitor types and programs, while also generating insights into donor behavior and targeting. These efforts will support more informed, data-driven decisions to increase participation, engagement, and overall impact.

Tools and Data Sources

Resource	Implementation
VERSAI	Dashboard creation
SQL	Data extraction and queries
Apache Superset	Data access

Deliverables

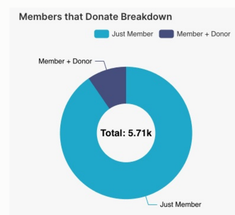
Visualization 1: Donor Insights Dashboard



Key Insight

This scatter plot displays individual donors (represented by each dot) as points based on the number of days since their last donation. The graph can be used to identify active vs inactive donors, understand patterns or behavior over time, and support target fundraising strategies.

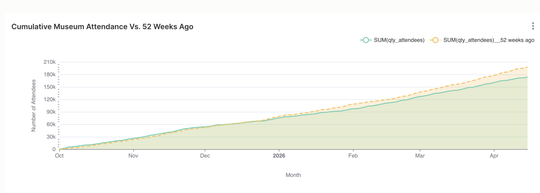
Visualization 2: Membership Dashboard



Key Insight

These figures show that membership sales are higher this year compared to last year across both front desk and online channels, with the front desk driving the majority of signups. However, while the overall member base is strong, only a small portion of members also donate, highlighting a key opportunity to improve donor conversion.

Visualization 3: Attendance Dashboard



Key Insight

This graph shows that cumulative attendance is steadily increasing this year but consistently trails last year's pace, indicating a sustained performance gap rather than a short-term fluctuation. This suggests underlying differences in demand or engagement, highlighting a need for targeted strategies to accelerate attendance growth and close the gap.

Methods

What was our process?

- Conducted exploratory analysis to understand museum data
- Built structured datasets and transformations using SQL
- Designed interactive dashboards in Apache Superset
- Iterated on dashboard design and functionality based on client feedback

What feedback did we receive? What did we learn?

- Year-over-year trends provide more meaningful insights than daily fluctuations
- Program-level segmentation is critical for decision-making
- Clear, intuitive visualizations significantly improve usability and adoption
- Data accuracy, consistency, and validation are essential for reliable insights

Impact

What were the outcomes?

- Developed three interactive dashboards (attendance, membership, donor) to centralize key insights
- Identified actionable trends in visitor engagement, participation, and donor behavior
- Increased visibility and accessibility of museum data for more effective decision-making

How will this impact the community?

- Helps the museum to better serve its visitor base
- Supports outreach to underserved communities
- Improves program planning and accessibility
- Strengthens long-term fundraising efforts

Conclusion

This project demonstrates how data visualization can transform raw data into meaningful insights. By utilizing these dashboards, the museum can make smarter decisions, improve engagement, and better serve the community. These tools provide a foundation for continuous learning, growth, and long-term impact.

Visit the Museum & View the Full Report

Ann Arbor Hands-On Museum & Donate

User Documentation Guide